



Taller práctico de Git

Crear una simulación de “página web” para una cafetería llamada Café Aurora.

No usaremos código complejo.

El objetivo es aprender Git paso a paso como en un entorno profesional.

Usaremos:

- ramas
- commits
- merge
- stash
- tags
- remotos
- conflictos
- restauración
- historial
- rebase
- etc.

Caso práctico

Tiene esta carpeta: `cafe-aurora/`

Dentro habrá archivos simples:

```
index.html  
menu.txt  
contacto.txt  
README.md
```

PARTE 1 Crear repositorio

1) git init

Inicializa Git.

```
git init
```

Resultado:

```
Initialized empty Git repository
```



```
Usuario@DESKTOP MINGW64 ~ (master)
$ git init
Reinitialized existing Git repository in C:/Users/Usuario/.git/

Usuario@DESKTOP MINGW64 ~ (master)
$ rm -ff .git
rm: cannot remove '.': Is a directory

Usuario@DESKTOP MINGW64 ~ (master)
$ rm -rf .git

Usuario@DESKTOP MINGW64 ~
$ mkdir cafe-aurora

Usuario@DESKTOP MINGW64 ~
$ cd cafe-aurora

Usuario@DESKTOP MINGW64 ~/cafe-aurora
$ git init
Initialized empty Git repository in C:/Users/Usuario/cafe-aurora/.git/
```

2) git status

Ver estado actual.

```
git status
```

```
Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git status
On branch master

No commits yet

nothing to commit (create/copy files and use "git add" to track)
```

3) git config

Configurar usuario.

```
git config --global user.name "Juan"
git config --global user.email "juan@email.com"
```

Ver configuración:

```
git config --list
```



```
Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git config --global user.name "eimy vargas"
gigi
Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git config --global user.email "eimyvargas170@gmail.com"

Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=eimy vargas
user.email=eimyvargas170@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
```

PARTE 2 Primer commit

Crea:

README.md

Contenido:

Proyecto cafetería Cafe Aurora

4) git add

git add [README.md](#)



```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~ (master)
$ README.md
bash: README.md: command not found

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~ (master)
$ touch README.md

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~ (master)
$ git add README.md

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~ (master)
$ cd ~/Documents

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ mkdir cafeaurora
```

5) git status

```
git status
```

Verás:

```
Changes to be committed
```

6) git commit

```
git commit -m "Primer commit"
```

7) git log

```
git log
```



```
Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ touch README.md

Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        README.md

nothing added to commit but untracked files present (use "git add" to track)

Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git add README.md

Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   README.md

Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git commit -m "Primer commit"
[master (root-commit) 571aa61] Primer commit
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 README.md

Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git log
commit 571aa61ac001bc7b37421d0713badab74777fab0 (HEAD -> master)
Author: eimy vargas <eimyvargas170@gmail.com>
Date:   Fri May 29 12:24:10 2026 -0500

    Primer commit

Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
```

8) git log --oneline

git log --oneline

```
Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git log --oneline
571aa61 (HEAD -> master) Primer commit
```



PARTE 3 Trabajar con archivos

Crea:

```
menu.txt
```

9) git add .

```
git add .
```

10) git diff

Modifica menu.txt.

Luego:

```
git diff
```

Muestra cambios no staged.

11) git commit

```
git commit -m "Agrega menú inicial"
```



```
Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ touch menu.txt

Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        menu.txt

nothing added to commit but untracked files present (use "git add" to track)

Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git add .

Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git diff

Usuario@DESKTOP MINGW64 ~/cafe-aurora (master)
$ git commit -m "Agrega menú inicial"
[master f78b4ab] Agregá menú inicial
1 file changed, 0 insertions(+), 0 deletions(-)
create mode 100644 menu.txt
```

PARTE 4 Branches

12) git branch

```
git branch
```

13) git branch feature/contacto

```
git branch feature/contacto
```

14) git checkout

```
git checkout feature/contacto
```

15) git switch

Volver a main:

```
git switch main
```

16) git checkout -b



Crear y cambiar:

```
git checkout -b feature/redes-sociales
```

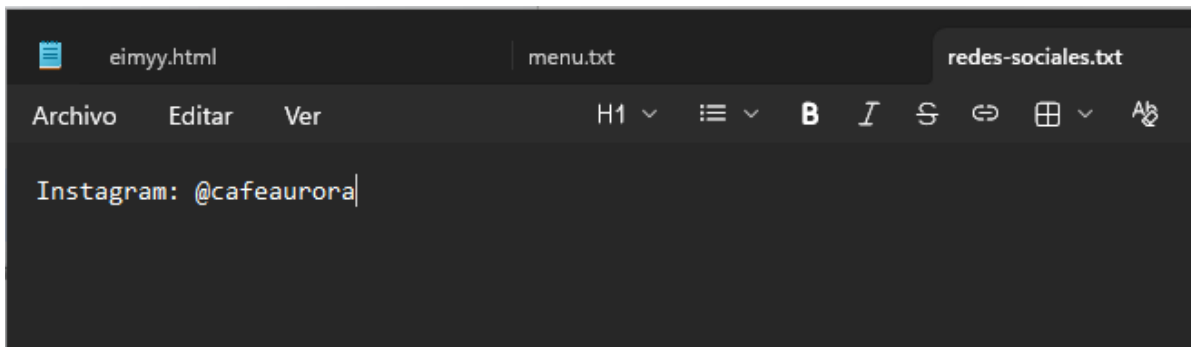
```
Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (master)
$ git checkout -b feature/redes-sociales
Switched to a new branch 'feature/redes-sociales'

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (feature/redes-sociales)
$ |
```

PARTE 5 Merge

Agrega:

```
instagram: @cafeaurora
```



Commit:

```
git add .
git commit -m "Agrega redes sociales"
```

```
Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (feature/redes-sociales)
$ git add .
warning: could not open directory 'Documents/Mi música/': Permission denied
warning: could not open directory 'Documents/Mis imágenes/': Permission denied
warning: could not open directory 'Documents/Mis videos/': Permission denied

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (feature/redes-sociales)
$ git commit -m "agrega redes sociales"
[feature/redes-sociales 37cf629] agrega redes sociales
2 files changed, 2 insertions(+)
create mode 100644 Documents/redes-sociales.txt

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (feature/redes-sociales)
$ |
```

17) git merge

Volver a main:



```
git switch main
```

Fusionar:

```
git merge feature/redes-sociales
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git merge feature/redes-sociales
Updating aa0e830..37cf629
Fast-forward
 Documents/menu.txt      | 1 +
 Documents/redes-sociales.txt | 1 +
 2 files changed, 2 insertions(+)
 create mode 100644 Documents/redes-sociales.txt

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$
```

PARTE 6 Eliminar y mover archivos

18) git rm

```
git rm contacto.txt
```

19) git mv

```
git mv menu.txt menu-cafeteria.txt
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git mv menu.txt menu-cafeteria.txt
```

20) git commit

```
git commit -m "Reorganiza archivos"
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git commit -m "Reorganiza los archivos"
[master 4352da4] Reorganiza los archivos
 1 file changed, 0 insertions(+), 0 deletions(-)
 rename Documents/{menu.txt => menu-cafeteria.txt} (100%)
```

PARTE 7 Restaurar cambios

Modifica README.

21) git restore



```
git restore README.md
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git restore README.md
error: pathspec 'README.md' did not match any file(s) known to git

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ |
```

Descarta cambios.

22) git restore --staged

```
git restore --staged README.md
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git restore --staged README.md
error: pathspec 'README.md' did not match any file(s) known to git
```

Quita del staging.

PARTE 8 Trabajar con remoto (GitHub)

23) git remote add

```
git remote add origin https://github.com/juan/cafe-aurora.git
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git remote add origin https://github.com/eimyvargas170-ai/cafe-aurora.git
```

24) git remote -v

```
git remote -v
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git remote -v
origin https://github.com/eimyvargas170-ai/cafe-aurora.git (fetch)
origin https://github.com/eimyvargas170-ai/cafe-aurora.git (push)
```

25) git push

```
git push -u origin main
```



```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git remote -v
origin https://github.com/eimyvargas170-ai/cafe-aurora.git (fetch)
origin https://github.com/eimyvargas170-ai/cafe-aurora.git (push)

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git push -u origin master
info: please complete authentication in your browser...
Enumerating objects: 21, done.
Counting objects: 100% (21/21), done.
Delta compression using up to 8 threads
Compressing objects: 100% (17/17), done.
Writing objects: 100% (21/21), 7.30 KiB | 2.43 MiB/s, done.
Total 21 (delta 3), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (3/3), done.
To https://github.com/eimyvargas170-ai/cafe-aurora.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ |
```

26) git pull

git pull

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git pull
Already up to date.
```

27) git fetch

git fetch

28) git clone

Otro desarrollador haría:

git clone <https://github.com/juan/cafe-aurora.git>

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git clone https://github.com/eimyvargas170-ai/cafe-aurora.git
Cloning into 'cafe-aurora'...
remote: Enumerating objects: 21, done.
remote: Counting objects: 100% (21/21), done.
remote: Compressing objects: 100% (14/14), done.
remote: Total 21 (delta 3), reused 21 (delta 3), pack-reused 0 (from 0)
Receiving objects: 100% (21/21), 7.30 KiB | 3.65 MiB/s, done.
Resolving deltas: 100% (3/3), done.

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
```

git

PARTE 9 Stash



Estás trabajando y necesitas cambiar de rama rápido.

29) git stash

git stash

```
Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (master)
$ git stash
Saved working directory and index state WIP on master: 4352da4 Reorganiza los archivos
Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (master)
```

Guarda temporalmente cambios.

30) git stash list

git stash list

```
Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (master)
$ git stash list
stash@{0}: WIP on master: 4352da4 Reorganiza los archivos
```

31) git stash pop

git stash pop



```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git stash pop
warning: could not open directory 'Configuración local/': Permission denied
warning: could not open directory 'Cookies/': Permission denied
warning: could not open directory 'Datos de programa/': Permission denied
warning: could not open directory 'Documents/Mi música/': Permission denied
warning: could not open directory 'Documents/Mis imágenes/': Permission denied
warning: could not open directory 'Documents/Mis vídeos/': Permission denied
warning: could not open directory 'Entorno de red/': Permission denied
warning: could not open directory 'Impresoras/': Permission denied
warning: could not open directory 'Menú Inicio/': Permission denied
warning: could not open directory 'Mis documentos/': Permission denied
warning: could not open directory 'Plantillas/': Permission denied
warning: could not open directory 'Reciente/': Permission denied
warning: could not open directory 'SendTo/': Permission denied
On branch master
Your branch is up to date with 'origin/master'.

Changes not staged for commit:
  (use "git add/rm <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        deleted:    ../README.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        ../.bash_history
        ../.config/
        ../.git-for-windows-updater
        ../.gitconfig
        ../.lessht
        ../.m2/
        ../.nbi/
        ../.vscode-shared/
        ../.vscode/
        ../AppData/
        ../Application Data/
        ../Contacts/
        ../Desktop/
        README.txt
        cafe-aurora/
        ../Downloads/
        ../Favorites/
        ../IntelGraphicsProfiles/
        ../Links/
        ../Music/
        ../NTUSER.DAT
        ../NTUSER.DAT{d350bf1c-f04a-11ef-9e70-e86f38a40c6e}.TM.blf
        ../NTUSER.DAT{d350bf1c-f04a-11ef-9e70-e86f38a40c6e}.TMContainer00000000000000000001.regtrans-ms
        ../NTUSER.DAT{d350bf1c-f04a-11ef-9e70-e86f38a40c6e}.TMContainer00000000000000000002.regtrans-ms
        ../OneDrive/
        ../Pictures/
        ../Searches/
        ../Videos/
        ../alejandra/
        ../contra seg/
        ../ntuser.dat.LOG1
        ../ntuser.dat.LOG2
        ../ntuser.ini
        ../pseint/

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (fd47baa9c35c629a50d07370e4a84528eddaa338)
```

Recupera cambios.

PARTE 10 Historial y detalles

32) git show

git show



```
Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (master)
$ git show
commit 4352da448eaabb9e450e69253f0d2cac25eb3026 (HEAD -> master, origin/master, origin/HEAD)
Author: eimy vargas <eimyvargas170@gmail.com>
Date: Thu May 28 20:44:16 2026 -0500

    Reorganiza los archivos

diff --git a/Documents/menu.txt b/Documents/menu-cafeteria.txt
similarity index 100%
rename from Documents/menu.txt
rename to Documents/menu-cafeteria.txt

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (master)
$ |
```

33) git reflog

git reflog

```
Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (master)
$ git reflog
4352da4 (HEAD -> master, origin/master, origin/HEAD) HEAD@{0}: reset: moving to HEAD
4352da4 (HEAD -> master, origin/master, origin/HEAD) HEAD@{1}: commit: Reorganiza los archivos
37cf629 (feature/redes-sociales) HEAD@{2}: merge feature/redes-sociales: Fast-forward
aa0e830 (feature/contacto) HEAD@{3}: checkout: moving from feature/redes-sociales to master
37cf629 (feature/redes-sociales) HEAD@{4}: commit: agrega redes sociales
aa0e830 (feature/contacto) HEAD@{5}: checkout: moving from master to feature/redes-sociales
aa0e830 (feature/contacto) HEAD@{6}: checkout: moving from feature/contacto to master
aa0e830 (feature/contacto) HEAD@{7}: checkout: moving from master to feature/contacto
aa0e830 (feature/contacto) HEAD@{8}: commit: Agrega menú inicial
edd2b37 HEAD@{9}: commit (initial): primer commit

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (master)
$ |
```

Historial interno.

34) git blame

git blame [README.md](#)

```
Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (master)
$ git blame README.md
fatal: no such path 'Documents/README.md' in HEAD

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Documents (master)
$ |
```

Quién modificó cada línea.

PARTE 11 Tags y versiones

35) git tag

git tag v1.0



```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git tag v1.0
```

36) git tag -a

```
git tag -a v1.1 -m "Versión estable"
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git tag -a v1.1 -m "Version estable"
```

37) git push --tags

```
git push --tags
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 164 bytes | 164.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/eimyvargas170-ai/cafe-aurora.git
 * [new tag]          v1.0 -> v1.0
 * [new tag]          v1.1 -> v1.1
```

PARTE 12 Reset y revert

38) git reset

```
git reset HEAD README.md
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Documents (master)
$ git reset HEAD README.md
Unstaged changes after reset:
D    README.md
```

Quitar del stgaging.

39) git revert

```
git revert HEAD
```



```
Revert "Reorganiza los archivos"

This reverts commit 4352da448eaabb9e450e69253f0d2cac25eb3026.

# Please enter the commit message for your changes. Lines starting
# with '#' will be ignored, and an empty message aborts the commit.
#
# On branch master
# Your branch is up to date with 'origin/master'.
#
# Changes to be committed:
#   renamed:   Documents/menu-cafeteria.txt -> Documents/menu.txt
#
# Changes not staged for commit:
#   deleted:   README.md
#
# Untracked files:
#   .bash_history
#   .config/
#   .git-for-windows-updater
#   .gitconfig
#   .lessht
#   .m2/
#   .nbi/
#   .vscode-shared/
#   .vscode/
#   AppData/
#   Application Data/
#   Contacts/
#   Desktop/
#   Documents/README.txt
#   Documents/cafe-aurora/
#   Downloads/
#   Favorites/
#   IntelGraphicsProfiles/
#   Links/
#   Music/
#   NTUSER.DAT
#   NTUSER.DAT{d350bf1c-f04a-11ef-9e70-e86f38a40c6e}.TM.blf
#   NTUSER.DAT{d350bf1c-f04a-11ef-9e70-e86f38a40c6e}.TMContainer000000000000000001.regtrans-ms
#   NTUSER.DAT{d350bf1c-f04a-11ef-9e70-e86f38a40c6e}.TMContainer000000000000000002.regtrans-ms
#   OneDrive/
#   Pictures/
#   Searches/
#   Videos/
#   alejandra/
#   contra seg/
#   ntuser.dat.LOG1
#   ntuser.dat.LOG2
#   ntuser.ini
#   pseint/
#
~
~
~
~
~
```

Deshacer commit sin borrar historial.

PARTE 13 Rebase

40) git rebase

```
git rebase main
```




```
Aprendiz@DESKTOP-LPU1C05 MINGW64 ~ (master)
$ git rebase main
fatal: invalid upstream 'main'

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~ (master)
$ |
```

Reorganiza historial.

EXTRA Flujo profesional

```
git pull
git switch -c feature/menu
git add .
git commit -m "Actualiza menú"
git push origin feature/menu
```

```
Aprendiz@DESKTOP-LPU1C05 MINGW64 ~ (master)
$ git rebase main
fatal: invalid upstream 'main'

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~ (master)
$ git pull
git switch -c feature/menu
git add .
git commit -m "Actualiza menú"
git push origin feature/menu
Already up to date.
Switched to a new branch 'feature/menu'
warning: could not open directory 'AppData/Local/Archivos temporales de Internet/': Permission denied
warning: could not open directory 'AppData/Local/Datos de programa/': Permission denied
warning: could not open directory 'AppData/Local/Historial/': Permission denied
warning: could not open directory 'AppData/Local/Microsoft/Windows/NetCache/Content.IE5/': Permission denied
warning: could not open directory 'AppData/Local/Microsoft/Windows/NetCache/Content.IE5/': Permission denied
warning: could not open directory 'AppData/Local/Microsoft/Windows/Temporary Internet Files/': Permission denied
warning: could not open directory 'AppData/Roaming/Microsoft/Windows/Start Menu/Programas/': Permission denied
warning: could not open directory 'Configuración local/': Permission denied
warning: could not open directory 'Cookies/': Permission denied
warning: could not open directory 'Datos de programa/': Permission denied
warning: could not open directory 'Documents/Mis música/': Permission denied
warning: could not open directory 'Documents/Mis imágenes/': Permission denied
warning: could not open directory 'Documents/Mis videos/': Permission denied
warning: could not open directory 'Entorno de red/': Permission denied
warning: could not open directory 'Impresoras/': Permission denied
warning: could not open directory 'Menú Inicio/': Permission denied
warning: could not open directory 'Mis documentos/': Permission denied
warning: could not open directory 'Plantillas/': Permission denied
warning: could not open directory 'Reciente/': Permission denied
warning: could not open directory 'Sendto/': Permission denied
warning: in the working copy of '.bash_history', LF will be replaced by CRLF the
next time Git touches it
warning: in the working copy of '.config/git/config', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.git-for-windows-updater', LF will be replaced
by CRLF the next time Git touches it
warning: in the working copy of '.gitconfig', LF will be replaced by CRLF the ne
xt time Git touches it
warning: in the working copy of '.lessshst', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.m2/repository/com/fasterxml/jackson/core/jacks
on-annotation/2.19.0/jackson-annotation-2.19.0.pom', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.m2/repository/com/fasterxml/jackson/core/jacks
on-databind/2.19.0/jackson-databind-2.19.0.pom', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.m2/repository/com/fasterxml/jackson/datatype/j
ackson-datatype-jdk8/2.19.0/jackson-datatype-jdk8-2.19.0.pom', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.m2/repository/com/fasterxml/jackson/datatype/j
ackson-datatype-jsr310/2.19.0/jackson-datatype-jsr310-2.19.0.pom', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.m2/repository/com/fasterxml/jackson/jackson-base/2.19.0/jackson-base-2.19.0.pom', LF will be replaced by CRLF the next time Gi
t touches it
warning: in the working copy of '.m2/repository/com/fasterxml/jackson/jackson-bo
m/2.17.2/jackson-bom-2.17.2.pom', LF will be replaced by CRLF the next time Git
touches it
warning: in the working copy of '.m2/repository/com/fasterxml/jackson/jackson-bo
m/2.19.0/jackson-bom-2.19.0.pom', LF will be replaced by CRLF the next time Git
touches it
warning: in the working copy of '.m2/repository/com/fasterxml/jackson/jackson-pa
rent/2.17/jackson-parent-2.17.pom', LF will be replaced by CRLF the next time Gi
t touches it
warning: in the working copy of '.m2/repository/com/fasterxml/jackson/jackson-pa
rent/2.19/jackson-parent-2.19.pom', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.m2/repository/com/fasterxml/jackson/module/jac
kson-module-parameter-names/2.19.0/jackson-module-parameter-names-2.19.0.pom', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.m2/repository/com/fasterxml/jackson/module/jac
kson-modules-java8/2.19.0/jackson-modules-java8-2.19.0.pom', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.m2/repository/com/fasterxml/jackson/oss-parent/38/oss-
parent-38.pom', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.m2/repository/com/fasterxml/jackson/oss-parent/65/oss-
parent-65.pom', LF will be replaced by CRLF the next time Git touches it
warning: in the working copy of '.m2/repository/com/jayway/jsonpath/json-path/2.9.0/json-path-2.9.0.pom', LF will be replaced by CRLF the next time Git touches
```

RESUMEN VISUAL



Working Directory
↓
git add
↓
Staging Area
↓
git commit
↓
Repositorio local
↓
git push
↓
GitHub

LOS 40 COMANDOS APRENDIDOS

- 1 git init
- 2 git status
- 3 git config
- 4 git add
- 5 git commit
- 6 git log
- 7 git log --oneline
- 8 git diff
- 9 git branch
- 10 git checkout
- 11 git switch
- 12 git merge
- 13 git rm
- 14 git mv
- 15 git restore
- 16 git remote
- 17 git push
- 18 git pull
- 19 git fetch
- 20 git clone
- 21 git stash
- 22 git stash list
- 23 git stash pop
- 24 git show
- 25 git reflog
- 26 git blame
- 27 git tag
- 28 git tag -a
- 29 git push --tags



```
30 git reset
31 git revert
32 git rebase
33 git checkout -b
34 git remote add
35 git restore --staged
36 git branch feature/*
37 git switch main
38 git merge feature/*
39 git add .
40 git commit -m
```

¿Te atreves a realizar el siguiente reto?

Una **simulación real de empresa**, donde usaras:

- comandos básicos (add, commit, branch...)
- comandos intermedios (stash, rebase, reset...)
- comandos avanzados (reflog, bisect, cherry-pick...)
- inspección interna de Git
- herramientas de debugging
- configuración
- limpieza
- flujos de equipo

E-commerce Café Aurora pro

Trabaja en equipo con GitHub. Para desarrollar:

- tienda online
- pagos
- menú
- sistema de usuarios
- corrección de bugs

PARTE 1 CONFIGURACIÓN INICIAL (10 comandos)

```
git --version
git config --global user.name "Juan"
git config --global user.email "juan@email.com"
git config --list
git help
```



```
git help commit
git init
git clone https://github.com/cafe/aurora.git
git remote -v
git remote add origin URL
```

PARTE 2 INSPECCIÓN DEL PROYECTO (10 comandos)

```
git status
git status -s
git log
git log --oneline
git log --graph
git log --all
git diff
git diff HEAD
git diff --staged
git show HEAD
```

PARTE 3 WORKFLOW BÁSICO (10 comandos)

```
git add archivo.txt
git add .
git add -A
git commit -m "mensaje"
git commit -am "mensaje"
git rm archivo.txt
git mv viejo.txt nuevo.txt
git restore archivo.txt
git restore --staged archivo.txt
git clean -f
```

PARTE 4 RAMAS (12 comandos)

```
git branch
git branch feature/login
git branch -d feature/login
git branch -D feature/login
git checkout main
git checkout -b feature/login
git switch main
git switch -c feature/login
git merge feature/login
git merge --no-ff feature/login
git branch -m nuevo-nombre
git branch -r
```

PARTE 5 REBASE Y HISTORIAL (8 comandos)

```
git rebase main
git rebase -i HEAD~3
```



```
git rebase --abort
git rebase --continue
git cherry-pick abc1234
git revert HEAD
git reset HEAD~1
git reset --soft HEAD~1
```

PARTE 6 REMOTO (10 comandos)

```
git push
git push -u origin main
git push origin feature/login
git pull
git fetch
git fetch --all
git remote add origin URL
git remote remove origin
git remote show origin
git ls-remote
```

PARTE 7 STASH (6 comandos)

```
git stash
git stash list
git stash pop
git stash apply
git stash drop
git stash clear
```

PARTE 8 RECUPERACIÓN Y DEBUG (10 comandos)

```
git reflog
git reset --hard HEAD
git reset --mixed HEAD
git reset --soft HEAD
git fsck
git gc
git blame archivo.txt
git show commit-id
git bisect start
git bisect bad
git bisect Good
```

PARTE 9 VERSIONES (5 comandos)

```
git tag v1.0
git tag -a v1.1 -m "release"
git tag
git push --tags
git show v1.0
```

PARTE 10 LIMPIEZA (5 comandos)

```
git clean -n
git clean -f
git clean -fd
```



```
git prune  
git gc -aggressive
```

PARTE 11 UTILIDADES AVANZADAS (5 comandos)

```
git shortlog  
git archive  
git describe  
git check-ignore  
git update-index
```

PARTE 12 AYUDA Y CONTROL (5 comandos)

```
git help  
git help merge  
git help rebase  
git help stash  
git config --global -l
```

PARTE 13 FLUJO REAL DE TRABAJO (EQUIPO)

Ejemplo completo de trabajo profesional:

```
git pull  
git checkout -b feature/payments  
git add .  
git commit -m "Add payments system"  
git push origin feature/payments  
git fetch  
git rebase main  
git merge feature/payments  
git push
```

RESUMEN

En empresas se usa realmente esto:

20% de comandos hacen 80% del trabajo:

- git add
- git commit
- git push
- git pull
- git branch



- git checkout / switch
- git merge
- git stash
- git log
- git reset

IMPORTANTE

Muchos comandos:

- se usan solo en casos extremos (bisect, fsck, gc)
- o en debugging
- o en equipos grandes

CONCLUSIÓN

Git no se aprende por cantidad de comandos, sino por entender flujo:

Working Directory → Staging → Commit → Remote

Taller Práctico de Git

Proyecto: Café Aurora

“Aprender Git no es memorizar comandos, es entender cómo trabajan los equipos reales de desarrollo.”

Objetivo del Taller

Construiremos paso a paso el proyecto Café Aurora, una página web sencilla para una cafetería ficticia.

Durante el proceso aprenderás cómo se trabaja con Git en ambientes profesionales usando:

- Commits
- Ramas (branches)
- Merge
- Stash
- Rebase
- Tags



- GitHub y remotos
- Restauración de archivos
- Historial y debugging
- Resolución de conflictos
- Flujo de trabajo colaborativo

Al finalizar el taller ¿qué aprenderás realmente?

- Crear repositorios profesionales
- Controlar versiones de proyectos
- Trabajar en equipo usando GitHub
- Recuperar cambios perdidos
- Resolver errores comunes
- Entender el flujo real de desarrollo en empresas

Escenario del Proyecto

Supón que trabajas como desarrollador en la cafetería digital:

Café Aurora

Tu equipo necesita crear una pequeña web para mostrar:

- menú
- contacto
- redes sociales
- información del negocio

Estructura inicial

```
cafe-aurora/  
├── index.html  
├── menu.txt  
├── contacto.txt  
└── README.md
```

PARTE 1 Crear el repositorio

1. Inicializar Git



```
git init
```

Convierte la carpeta en un repositorio Git.

Resultado esperado:

```
Initialized empty Git repository
```

2. Verificar estado

```
git status
```

Muestra archivos modificados, nuevos o pendientes.

3. Configurar usuario

```
git config --global user.name "Juan"  
git config --global user.email "juan@email.com"
```

Ver configuración:

```
git config --list
```

Git guardará quién realiza cada cambio.

PARTE 2 Primer Commit

Crear README.md

Ingresa el siguiente contenido:

```
Proyecto cafetería Cafe Aurora
```

4. Agregar archivo al staging

```
git add README.md
```

Prepara archivos para el commit.

5. Verificar cambios preparados



```
git status
```

Verá:

```
Changes to be committed
```

6. Crear primer commit

```
git commit -m "Primer commit"
```

Guarda una versión oficial del proyecto.

7. Ver historial

```
git log
```

8. Historial resumido

```
git log --oneline
```

PARTE 3 Trabajar con archivos

Crear archivo

```
menu.txt
```

9. Agregar todos los archivos

```
git add .
```

El punto significa: “todo”.

10. Ver diferencias

Modifica `menu.txt` y ejecuta:

```
git diff
```

Muestra cambios aún no preparados.



11. Guardar cambios

```
git commit -m "Agrega menú inicial"
```

PARTE 4 Ramas (Branches)

12. Ver ramas

```
git branch
```

13. Crear rama

```
git branch feature/contacto
```

Las ramas permiten trabajar sin afectar la versión principal.

14. Cambiar de rama

```
git checkout feature/contacto
```

15. Volver a main

```
git switch main
```

16. Crear y cambiar en un solo comando

```
git checkout -b feature/redes-sociales
```

PARTE 5 Merge

Agregar:

```
instagram: @cafeaurora
```

Guardar cambios:

```
git add .  
git commit -m "Agrega redes sociales"
```

17. Fusionar ramas



Volver a main:

```
git switch main
```

Fusionar:

```
git merge feature/redes-sociales
```

Une el trabajo de dos ramas.

PARTE 6 Eliminar y mover archivos

18. Eliminar archivos

```
git rm contacto.txt
```

19. Renombrar archivos

```
git mv menu.txt menu-cafeteria.txt
```

20. Confirmar cambios

```
git commit -m "Reorganiza archivos"
```

PARTE 7 Restaurar cambios

Modifica README.md

21. Restaurar archivo

```
git restore README.md
```

Descarta cambios no guardados.

22. Quitar del staging

```
git restore --staged README.md
```

PARTE 8 Trabajar con GitHub



23. Conectar repositorio remoto

```
git remote add origin https://github.com/juan/cafe-aurora.git
```

24. Ver remotos

```
git remote -v
```

25. Subir proyecto

```
git push -u origin main
```

26. Descargar cambios

```
git pull
```

27. Actualizar referencias remotas

```
git fetch
```

28. Clonar proyecto

```
git clone https://github.com/juan/cafe-aurora.git
```

PARTE 9 Stash

Escenario real: debes cambiar rápido de rama sin perder trabajo.

29. Guardar cambios temporalmente

```
git stash
```

30. Ver stash guardados

```
git stash list
```

31. Recuperar cambios

```
git stash pop
```

PARTE 10 Historial y debugging



32. Ver detalles de commits

```
git show
```

33. Ver historial interno

```
git reflog
```

Uno de los comandos más poderosos de Git.

34. Ver quién modificó cada línea

```
git blame README.md
```

PARTE 11 Versiones y Releases

35. Crear versión

```
git tag v1.0
```

36. Crear tag anotado

```
git tag -a v1.1 -m "Versión estable"
```

37. Subir tags

```
git push --tags
```

PARTE 12 Reset y Revert

38. Quitar archivo del staging

```
git reset HEAD README.md
```

39. Deshacer commit sin borrar historial

```
git revert HEAD
```

Ideal para trabajo colaborativo.

PARTE 13 Rebase



40. Reorganizar historial

```
git rebase main
```

Mantiene un historial más limpio.

EXTRA Flujo profesional real

```
git pull
git switch -c feature/menu
git add .
git commit -m "Actualiza menú"
git push origin feature/menu
```

RESUMEN VISUAL DEL FLUJO GIT

```
Working Directory
  ↓
git add
  ↓
Staging Area
  ↓
git commit
  ↓
Repositorio local
  ↓
git push
  ↓
GitHub
```

LOS COMANDOS MÁS IMPORTANTES

El 80% del trabajo real usa:

```
git add
git commit
git push
git pull
git branch
git switch
git merge
git stash
git log
git reset
```

IMPORTANTE

Muchos comandos avanzados como `git bisect`, `git fsck`, `git gc`



Se usan principalmente para:

- debugging
- mantenimiento
- proyectos grandes
- recuperación avanzada

CONCLUSIÓN

Git no se aprende memorizando 100 comandos.

Git se aprende entendiendo el flujo:

```
Working Directory
      ↓
Staging
      ↓
Commit
      ↓
Remote Repository
```

RETO FINAL

Proyecto: E-commerce Café Aurora Pro

Trabajarás como si estuvieras en una empresa real usando:

- GitHub
- ramas por funcionalidades
- corrección de bugs
- trabajo en equipo
- rebase y merge
- debugging
- recuperación de errores
- versiones estables

Desafío Final

¿Serías capaz de:

- crear nuevas funcionalidades?
- resolver conflictos?
- recuperar commits eliminados?



- trabajar en equipo sin dañar el proyecto?
- desplegar versiones estables?

Meta del Taller

Al terminar este laboratorio podrás decir:

- “Sé usar Git en proyectos reales.”
- “Entiendo cómo trabajan los equipos de desarrollo.”
- “Puedo colaborar profesionalmente usando Git y GitHub.”

Objetivo del proyecto

Construir una tienda online para:

- vender café
- mostrar catálogo
- gestionar pedidos
- manejar usuarios
- integrar pagos
- publicar releases

ARQUITECTURA DEL REPOSITORIO

```
cafe-aurora/  
├── frontend/  
├── backend/  
├── docs/  
├── tests/  
├── .gitignore  
└── README.md
```

FASE 1 INICIO DEL PROYECTO

1. Clonar o iniciar repo

```
git clone https://github.com/cafe/aurora.git  
cd cafe-aurora
```

o si empiezas desde cero:

```
git init
```



2. Configuración del equipo

```
git config --global user.name "Developer 1"  
git config --global user.email "dev1@cafe.com"
```

3. Conectar remoto

```
git remote add origin https://github.com/cafe/aurora.git
```

FASE 2 PRIMER RELEASE BASE

4. Crear estructura inicial

```
git status
```

5. Agregar todo

```
git add .  
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora  
$ git init  
Initialized empty Git repository in C:/Users/Aprendiz/Desktop/cafe-aurora/.git/  
  
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora (master)  
$ git config --global user.name "eimy"  
  
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora (master)  
$ git config --global user.email "eimyvargas170@gmail.com"  
  
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora (master)  
$ git remote add origin https://github.com/eimyvargas170-ai/cafecito-aurora.git  
  
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora (master)  
$ git remote -v  
origin https://github.com/eimyvargas170-ai/cafecito-aurora.git (fetch)  
origin https://github.com/eimyvargas170-ai/cafecito-aurora.git (push)  
  
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora (master)  
$ git status  
On branch master  
  
No commits yet  
  
nothing to commit (create/copy files and use "git add" to track)  
  
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora (master)  
$ git add .
```

6. Primer commit

```
git commit -m "chore: estructura inicial del proyecto"
```



```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora (master)
$ git commit -m "chore: estructura inicial del proyecto"
[master (root-commit) 286413b] chore: estructura inicial del proyecto
1 file changed, 1 insertion(+)
 create mode 100644 README.md

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora (master)
$
```

7. Subir a GitHub

```
git push -u origin main
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora (master)
$ git push -u origin master
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 258 bytes | 258.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/eimyvargas170-ai/cafecito-aurora.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora (master)
$
```

FASE 3 DESARROLLO POR RAMAS

8. Crear rama de desarrollo

```
git checkout -b develop
```

9. Crear feature catálogo

```
git checkout -b feature/catalogo
```

10. Trabajar catálogo

Se agregan productos:

- café colombiano
- café espresso
- café premium

11. Guardar cambios



```
git add .
git commit -m "feat: catálogo de productos inicial"
```

```
Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Desktop/cafe-aurora (master)
$ git checkout -b develop
Switched to a new branch 'develop'

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Desktop/cafe-aurora (develop)
$ git checkout -b feature/catalogo
Switched to a new branch 'feature/catalogo'

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Desktop/cafe-aurora (feature/catalogo)
$ touch fronted/catalogo.txt
touch: cannot touch 'fronted/catalogo.txt': No such file or directory

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Desktop/cafe-aurora (feature/catalogo)
$ ls fronted
ls: cannot access 'fronted': No such file or directory

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Desktop/cafe-aurora (feature/catalogo)
$ mkdir fronted backend docs tests

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Desktop/cafe-aurora (feature/catalogo)
$ echo "cafe colombiano" >> fronted/catalogo.txt

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Desktop/cafe-aurora (feature/catalogo)
$ echo "cafe expresso" >> fronted/catalogo.txt

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Desktop/cafe-aurora (feature/catalogo)
$ ^[[200~ echo "cafe colombiano" >> fronted/catalogo.txt
bash: $'\E[200~': command not found

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Desktop/cafe-aurora (feature/catalogo)
$ echo "cafe premium" >> fronted/catalogo.txt

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Desktop/cafe-aurora (feature/catalogo)
$ git add .
warning: in the working copy of 'fronted/catalogo.txt', LF will be replaced by CRLF the next time Git touches it

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Desktop/cafe-aurora (feature/catalogo)
$ git commit -m "feat: catalogo de productos inicial"
[feature/catalogo 9d9e1e8] feat: catalogo de productos inicial
1 file changed, 3 insertions(+)
create mode 100644 fronted/catalogo.txt
```

12. Subir rama

```
git push origin feature/catalogo
```

```
Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Desktop/cafe-aurora (feature/catalogo)
$ git push origin feature/catalogo
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 376 bytes | 376.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'feature/catalogo' on GitHub by visiting:
remote:   https://github.com/eimyvargas170-ai/cafecito-aurora/pull/new/feature/catalogo
remote:
remote: To https://github.com/eimyvargas170-ai/cafecito-aurora.git
* [new branch]   feature/catalogo -> feature/catalogo

Aprendiz@DESKTOP-LPU1C05 MINGW64 ~/Desktop/cafe-aurora (feature/catalogo)
$
```

13. Merge a develop



```
git checkout develop
git merge feature/catalogo
```

```
Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora (develop)
$ git checkout develop
Already on 'develop'

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora (develop)
$ git merge feature/catalogo
Updating 286413b..9d9e1e8
Fast-forward
 fronted/catalogo.txt | 3 +++
 1 file changed, 3 insertions(+)
 create mode 100644 fronted/catalogo.txt

Aprendiz@DESKTOP-LPU1C0S MINGW64 ~/Desktop/cafe-aurora (develop)
$ |
```

FASE 4 SISTEMA DE PAGOS

14. Crear rama

```
git checkout -b feature/pagos
```

15. Cambios en pagos

Se agregan:

- PayPal
- tarjeta
- Stripe

```
Usuario@DESKTOP MINGW64 ~/Desktop/cafecito-aurora (master)
$ git fetch --all

Usuario@DESKTOP MINGW64 ~/Desktop/cafecito-aurora (master)
$ git checkout develop
branch 'develop' set up to track 'origin/develop'.
Switched to a new branch 'develop'

Usuario@DESKTOP MINGW64 ~/Desktop/cafecito-aurora (develop)
$ echo "tarjeta" >> backend/pagos.txt

Usuario@DESKTOP MINGW64 ~/Desktop/cafecito-aurora (develop)
$ echo "stripe" >> backend/pagos.txt
```

- el comando “git checkout -b feature/pagos” lo había hecho en el ambiente de formación, al llegar a la casa clone el repositorio para no empezar de cero, es por eso en la captura no se aprecia el comando



16. Guardar cambios

```
git add .
git commit -m "feat: integración de pagos"
```

```
Usuario@DESKTOP MINGW64 ~/Desktop/cafecito-aurora (develop)
$ git add .
warning: in the working copy of 'backend/pagos.txt', LF will be replaced by CRLF the next time Git touches it
```

```
Usuario@DESKTOP MINGW64 ~/Desktop/cafecito-aurora (develop)
$ git commit -m "feat: integracion de pagos"
[develop 0df5f78] feat: integracion de pagos
1 file changed, 2 insertions(+)
```

17. Subir rama

```
git push origin feature/pagos
```

```
Usuario@DESKTOP MINGW64 ~/Desktop/cafecito-aurora (develop)
$ git push origin feature/pagos
error: src refspec feature/pagos does not match any
error: failed to push some refs to 'https://github.com/eimyvargas170-ai/cafecito-aurora.git'
```

```
Usuario@DESKTOP MINGW64 ~/Desktop/cafecito-aurora (develop)
$ git push origin develop
Enumerating objects: 7, done.
Counting objects: 100% (7/7), done.
Delta compression using up to 4 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (4/4), 386 bytes | 386.00 KiB/s, done.
Total 4 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/eimyvargas170-ai/cafecito-aurora.git
be193a7..0df5f78 develop -> develop
```

FASE 5 CONFLICTO REAL (IMPORTANTE)

Dos personas modifican el mismo archivo:

checkout.html

18. Merge conflict

```
git checkout develop
git merge feature/pagos
```



Conflicto aparece.

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (feature/pagos)
$ git checkout develop
Switched to branch 'develop'
Your branch is ahead of 'origin/develop' by 3 commits.
  (use "git push" to publish your local commits)

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (develop)
$ git merge feature/pagos
Auto-merging frontend/checkout.html
CONFLICT (content): Merge conflict in frontend/checkout.html
Automatic merge failed; fix conflicts and then commit the result.

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (develop|MERGING)
$ |
```

19. Ver estado

git status

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (develop|MERGING)
$ git status
On branch develop
Your branch is ahead of 'origin/develop' by 3 commits.
  (use "git push" to publish your local commits)

You have unmerged paths.
  (fix conflicts and run "git commit")
  (use "git merge --abort" to abort the merge)

Unmerged paths:
  (use "git add <file>..." to mark resolution)
    both modified:   frontend/checkout.html

no changes added to commit (use "git add" and/or "git commit -a")

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (develop|MERGING)
$ |
```

20. Resolver conflicto manualmente



Editas archivo y decides versión correcta.

```
<<<<<< HEAD
<h1>Checkout develop</h1>
<h1>diseño de pagina de pagos</h1>
=====
<h1>diseño de la pagina de pagos</h1>
>>>>>> feature/pagos
```

```
<h1>Checkout develop</h1>
<h1>diseño de la pagina de pagos</h1>
```

21. Marcar como resuelto

```
git add checkout.html
git commit -m "fix: resolve merge conflict en pagos"
```

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (develop|MERGING)
$ git add frontend/checkout.html

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (develop|MERGING)
$ git commit -m "chore: resolve merge conflict en pagos"
[develop 18ee3d6] chore: resolve merge conflict en pagos

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (develop)
```




FASE 6 RELEASE

22. Crear rama release

```
git checkout -b release/v1.0
```

23. Ajustes finales

- bugs menores
- UI
- performance

24. Commit final

```
git commit -am "release: versión 1.0 lista"
```

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (release/v1.0)
$ git commit -am "release: versión 1.0 lista"
[release/v1.0 8619afe] release: versión 1.0 lista
1 file changed, 1 insertion(+)
```

25. Merge a main

```
git checkout main
git merge release/v1.0
```

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (develop)
$ git checkout master
Switched to branch 'master'

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master)
$ git merge release/v1.0
Already up to date.

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master)
$ |
```

26. Tag de versión

```
git tag -a v1.0 -m "Primera versión estable"
```

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master)
$ git tag -a v1.0 -m "primera version estable"
```

27. Subir todo



```
git push origin main
git push --tags
```

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master)
$ git push origin master
Enumerating objects: 31, done.
Counting objects: 100% (31/31), done.
Delta compression using up to 4 threads
Compressing objects: 100% (27/27), done.
Writing objects: 100% (28/28), 2.47 KiB | 316.00 KiB/s, done.
Total 28 (delta 11), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (11/11), completed with 1 local object.
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:   https://github.com/juliansito22acosta-svg/cafe-auroras/pull/new/master
remote:
To https://github.com/juliansito22acosta-svg/cafe-auroras.git
 * [new branch]      master -> master

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master)
$ git push --tags
Enumerating objects: 1, done.
Counting objects: 100% (1/1), done.
Writing objects: 100% (1/1), 167 bytes | 167.00 KiB/s, done.
Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/juliansito22acosta-svg/cafe-auroras.git
 * [new tag]         v1.0 -> v1.0
```

FASE 7 TRABAJO DE EMERGENCIA (HOTFIX)

28. Bug crítico en pagos

```
git checkout -b hotfix/pagos-error
```

29. Fix rápido

```
git commit -am "hotfix: error crítico en checkout"
```

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master)
$ git checkout -b hotfix/pagos-error
Switched to a new branch 'hotfix/pagos-error'

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (hotfix/pagos-error)
$ git commit -am "hotfix: error crítico en checkout"
On branch hotfix/pagos-error
nothing to commit, working tree clean
```

30. Merge inmediato

```
git checkout main
git merge hotfix/pagos-error
```



```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (hotfix/pagos-error)
$ git checkout master
Switched to branch 'master'

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master)
$ git merge hotfix/pagos-error
Already up to date.
```

FASE 8 HISTORIAL Y CONTROL

31. Ver historial

git log --oneline --graph

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master)
$ git log --oneline --graph
* 8619afe (HEAD -> master, tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop) release: versión 1.0 lista
* 18ee3d6 chore: resolve merge conflict en pagos
* 5a75c8a (feature/pagos) feat: diseño alternativo de pagos
* 65b349c feat: diseño de pago alternativo
* b8888ac feat: version eimy
* bd435fc feat: version eimy
* 3ee1181 feat: holaaa, diseño de checkout eimy
* 1917da4 (origin/develop, origin/HEAD) feat: checkout base
* 23fe027 feat: integracion de pagos
* 7cfdc91 feat: catalogo de productos inicial
* 686e621 chore: estructura inicial del proyecto
```

32. Ver cambios de un commit

git show HEAD

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master)
$ git show HEAD
commit 8619afe73e1e1a1c5215182f702376d774c12a7 (HEAD -> master, tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop)
Author: eimy vargas <eimyvargas170@gmail.com>
Date: Sun May 31 17:31:14 2026 -0500

    release: versión 1.0 lista

diff --git a/frontend/checkout.html b/frontend/checkout.html
index f0de5dc..66eb6f8 100644
--- a/frontend/checkout.html
+++ b/frontend/checkout.html
@@ -1,3 +1,4 @@

<h1>Checkout develop</h1>
<h1>diseño de la pagina de pagos</h1>
+<span>Version 1.0 lista</span>
\ No newline at end of file
```

33. Quién modificó qué

git blame backend/[payments.js](#)

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master)
$ git blame frontend/checkout.html
5a75c8a9 (eimy vargas 2026-05-31 17:08:28 -0500 1)
1917da42 (julian acosta 2026-05-30 20:38:04 -0500 2) <h1>Checkout develop</h1>
18ee3d63 (eimy vargas 2026-05-31 17:24:36 -0500 3) <h1>diseño de la pagina de pagos</h1>
c05f4967 (santiago ayala 2026-05-31 18:20:11 -0500 4) <span>Version 1.0 lista</span>
c05f4967 (santiago ayala 2026-05-31 18:20:11 -0500 5) <!--cambio de julian -->
c05f4967 (santiago ayala 2026-05-31 18:20:11 -0500 6) <!--ajustes finales de santiago -->
```

34. Recuperar estado anterior



git reflog

```
c05f496 (HEAD -> master) HEAD@{0}: commit: fix: revision de diseño final
8619afe (tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop) HEAD@{1}: checkout: moving from hotfix/pagos-error to master
8619afe (tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop) HEAD@{2}: checkout: moving from master to hotfix/pagos-error
8619afe (tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop) HEAD@{3}: checkout: moving from develop to master
8619afe (tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop) HEAD@{4}: merge release/v1.0: Fast-forward
18ee3d6 HEAD@{5}: checkout: moving from master to develop
8619afe (tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop) HEAD@{6}: checkout: moving from release/v1.0 to master
8619afe (tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop) HEAD@{7}: commit: release: versión 1.0 lista
18ee3d6 HEAD@{8}: checkout: moving from develop to release/v1.0
18ee3d6 HEAD@{9}: commit (merge): chore: resolve merge conflict in pagos
b8888ac HEAD@{10}: checkout: moving from feature/pagos to develop
5a75c8a (feature/pagos) HEAD@{11}: commit: feat: diseño alternativo de pagos
65b349c HEAD@{12}: checkout: moving from develop to feature/pagos
b8888ac HEAD@{13}: commit: feat: version eimy
bd435fc HEAD@{14}: commit: feat: version eimy
3ee1181 HEAD@{15}: reset: moving to HEAD-1
65b349c HEAD@{16}: merge feature/pagos: Fast-forward
v...skipping...
c05f496 (HEAD -> master) HEAD@{0}: commit: fix: revision de diseño final
8619afe (tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop) HEAD@{1}: checkout: moving from hotfix/pagos-error to master
8619afe (tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop) HEAD@{2}: checkout: moving from master to hotfix/pagos-error
8619afe (tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop) HEAD@{3}: checkout: moving from develop to master
8619afe (tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop) HEAD@{4}: merge release/v1.0: Fast-forward
18ee3d6 HEAD@{5}: checkout: moving from master to develop
8619afe (tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop) HEAD@{6}: checkout: moving from release/v1.0 to master
8619afe (tag: v1.0, origin/master, release/v1.0, hotfix/pagos-error, develop) HEAD@{7}: commit: release: versión 1.0 lista
18ee3d6 HEAD@{8}: checkout: moving from develop to release/v1.0
18ee3d6 HEAD@{9}: commit (merge): chore: resolve merge conflict in pagos
b8888ac HEAD@{10}: checkout: moving from feature/pagos to develop
5a75c8a (feature/pagos) HEAD@{11}: commit: feat: diseño alternativo de pagos
65b349c HEAD@{12}: checkout: moving from develop to feature/pagos
b8888ac HEAD@{13}: commit: feat: version eimy
bd435fc HEAD@{14}: commit: feat: version eimy
3ee1181 HEAD@{15}: reset: moving to HEAD-1
65b349c HEAD@{16}: merge feature/pagos: Fast-forward
3ee1181 HEAD@{17}: checkout: moving from feature/pagos to develop
65b349c HEAD@{18}: commit: feat: diseño de pago alternativo
3ee1181 HEAD@{19}: checkout: moving from develop to feature/pagos
3ee1181 HEAD@{20}: commit: feat: holaaa, diseño de checkout eimy
1917da4 (origin/develop, origin/HEAD) HEAD@{21}: checkout: moving from develop to develop
1917da4 (origin/develop, origin/HEAD) HEAD@{22}: clone: from https://github.com/juliansito22acosta-svg/cafe-auroras.git
```

Activar Windo
Ve a Configuración

FASE 9 STASH (trabajo urgente)

35. Guardar cambios temporales

git stash

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master)
$ git stash
Saved working directory and index state WIP on master: c05f496 fix: revision de
diseño final
```

36. Volver cambios

git stash pop

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master)
$ git stash pop
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   frontend/checkout.html

no changes added to commit (use "git add" and/or "git commit -a")
Dropped refs/stash@{0} (e10f6d02fe138e5fb9e79a32d38992610e4bafa2)
```



FASE 10 REBASE (historial limpio)

37. Reorganizar commits

```
git rebase main
```

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master)
$ git rebase master
Current branch master is up to date.
```

38. Rebase interactivo

```
git rebase -i HEAD~3
```

```
MINGW64:/c/Users/Usuario/Desktop/proyecto-grupo
pick 8619afe # release: versión 1.0 lista
pick c05f496 # fix: revision de diseño final
pick 30dae9c # chore: preparar entorno para rebase

# Rebase 18ee3d6..30dae9c onto 18ee3d6 (3 commands)
#
# Commands:
# p, pick <commit> = use commit
# r, reword <commit> = use commit, but edit the commit message
# e, edit <commit> = use commit, but stop for amending
# s, squash <commit> = use commit, but meld into previous commit
# f, fixup [-C | -c] <commit> = like "squash" but keep only the previous
#                               commit's log message, unless -C is used, in which case
#                               keep only this commit's message; -c is same as -C but
#                               opens the editor
# x, exec <command> = run command (the rest of the line) using shell
# b, break = stop here (continue rebase later with 'git rebase --continue')
# d, drop <commit> = remove commit
# l, label <label> = label current HEAD with a name
# t, reset <label> = reset HEAD to a label
# m, merge [-C <commit> | -c <commit>] <label> [# <oneline>]
#       create a merge commit using the original merge commit's
#       message (or the merge commit's message if -C is used)
.git/rebase-merge/git-rebase-todo [unix] (18:31 31/05/2026) 1,1 Top
<proyecto-grupo/.git/rebase-merge/git-rebase-todo" [unix] 34L, 1621B
```

FASE 11 DEBUG AVANZADO

39. Buscar bug

```
git bisect start
git bisect bad
git bisect good v0.9
```



```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master|BISECTING)
$ git bisect start
Already on 'master'
status: waiting for both good and bad commits

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master|BISECTING)
$ git bisect bad
status: waiting for good commit(s), bad commit known

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master|BISECTING)
$ git bisect good v0.9
error: Bad rev input: v0.9

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo (master|BISECTING)
$ git bisect good v1.0
Bisecting: 0 revisions left to test after this (roughly 0 steps)
[c05f49679422929e437ee0b105bd57bd413f9325] fix: revision de diseño final

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo ((c05f496...)|BISECTING)
$
```

40. Limpiar repo

```
git gc
git fsck
```

```
Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo ((c05f496...)|BISECTING)
$ git gc
Enumerating objects: 51, done.
Counting objects: 100% (51/51), done.
Delta compression using up to 4 threads
Compressing objects: 100% (42/42), done.
Writing objects: 100% (51/51), done.
Total 51 (delta 18), reused 13 (delta 3), pack-reused 0 (from 0)
Enumerating cruft objects: 5, done.
Traversing cruft objects: 8, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 4 threads
Compressing objects: 100% (5/5), done.
Writing objects: 100% (5/5), done.
Total 5 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)

Usuario@DESKTOP MINGW64 ~/Desktop/proyecto-grupo ((c05f496...)|BISECTING)
$ git fsck
Checking ref database: 100% (1/1), done.
Checking object directories: 100% (256/256), done.
Checking objects: 100% (56/56), done.
dangling commit e10f6d02fe138e5fb9e79a32d38992610e4bafa2
dangling tree f300131cb26127abc5837fb5f7b7ead258e95dca
Verifying commits in commit graph: 100% (13/13), done.
```

FLUJO FINAL DEL PROYECTO

```
feature → develop → release → main → tag
    ↓           ↓           ↓           ↓
  testing   staging   producción   versión
```



LO QUE APRENDISTE REALIZANDO EL ANTERIOR PROYECTO

Flujo real de empresa

- feature branches
- develop branch
- release branch
- hotfix branch

Trabajo en equipo

- merges
- conflictos
- rebase

Producción real

- tags
- versiones
- hotfix

Control total

- logs
- reflog
- blame
- bisect